MATHS 7105 Data Literacy

Assignment 1

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DATA ANALYSIS IN SUPPORT OF PARTY B

Indicator 1:

Economic Mismanagement and Declining Happiness under Opponents' Rule

The indicators used are the GDP growth rate vs Happiness survey (1-10) over the years. These indicators aim to show the citizen's happiness rate (measured by happiness survey rating)over the years with the correlation of economic growth(measured by GDP growth rate). The goal is to show how the economic mismanagement of Party A led to the decline of citizens' happiness.

There are no specific calculations made, just the GDP growth rate and Happiness survey have been analyzed from the given dataset.

GDP sores while citizens' Happiness plummets.

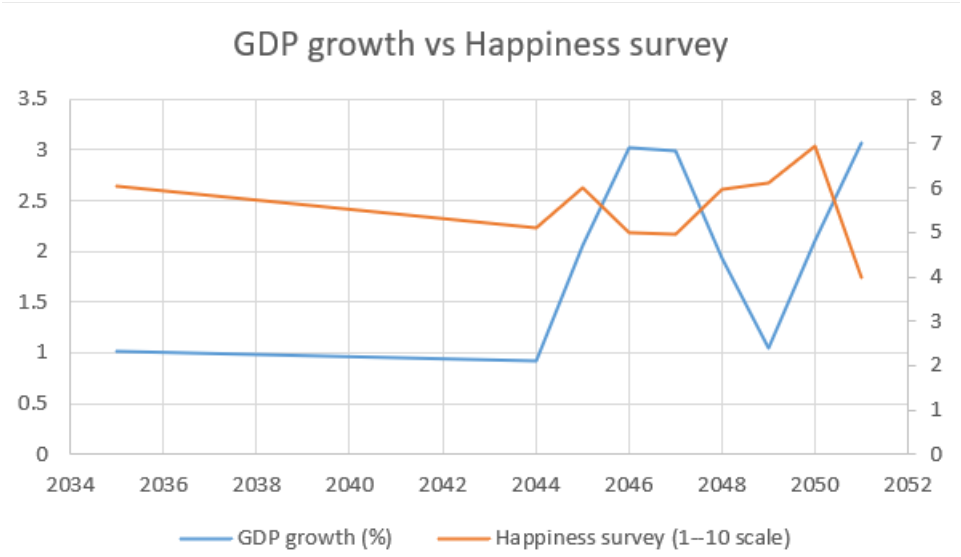


Fig.1 Happiness survey vs GDP growth.

The above graph shows the fluctuations in the GDP growth rate and the happiness survey of the citizens conducted over the years under the tenure of Party A. As seen during the years 2035 the GDP growth stood almost still while the happiness ratio was 6.05. In 2046 there are differences seen in the pattern of the GDP growth and the

Happiness survey. As the GDP is seen rising in 2046 it is reaching its peak , and despite of this high rise the data shows that the happiness rate is going down.

The GDP growth was relatively high the following years ranging from 2.99% to 3.07% while the happiness rating were decreasing continuously reaching its lowest in the year 2051.

This analysis highlights the party A's failure in boosting the peoples happiness , but instead they were more focused on boosting the GDP rate.

Indicator 2 :

The indicators used for this are the Education expenditure and the Unemployment rate in the country over the years under the Party A tenure.

The link between these two is that if more people are educated the more people will be competent to get a job.Analysing the link between unemployment rate and education expenditure spent it shows us that the government did raise money for it, but despite of increasing education expenditure, the unemployment rate is seen to be increasing as well. This questions that the money actually spent by Party A for education expenditure in according to reduce employment rate, where is all the money going?

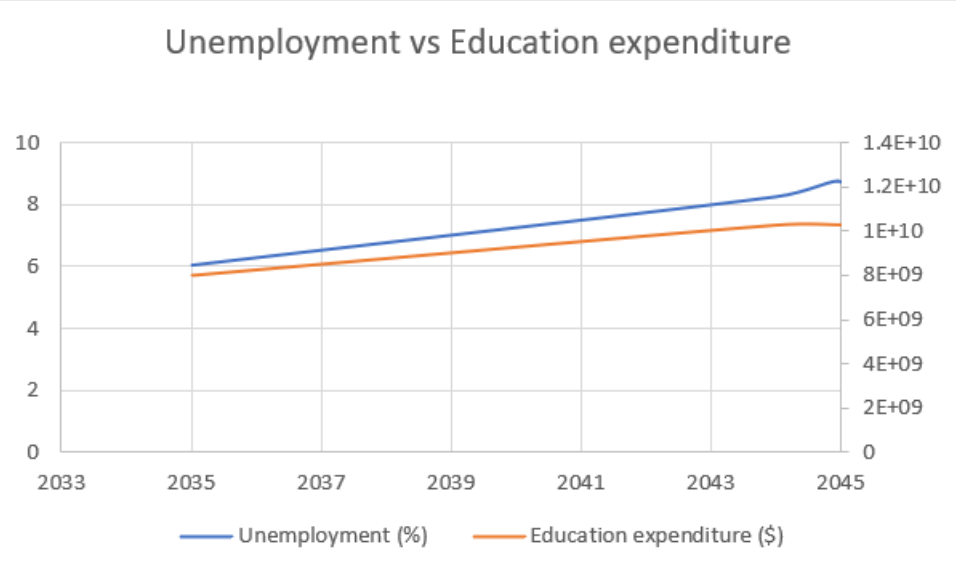


Fig.2 Unemployment [a1895211] vs Education expenditure.

Education Expenditure: A Failed Investment in Lowering Unemployment

Indicator 3 :

The indicators used are population figures and budget surplus values for the years of Party A's ruling period. The data exposes that even though the population growth was steady the budget surplus consistently plummeted. This shows the mismanagement of using the citizen's hard-earned money. This also shows how the opposition failed to introduce any proper budget policies, which also means that party A failed to adapt to the population change and ended up causing financial problems.

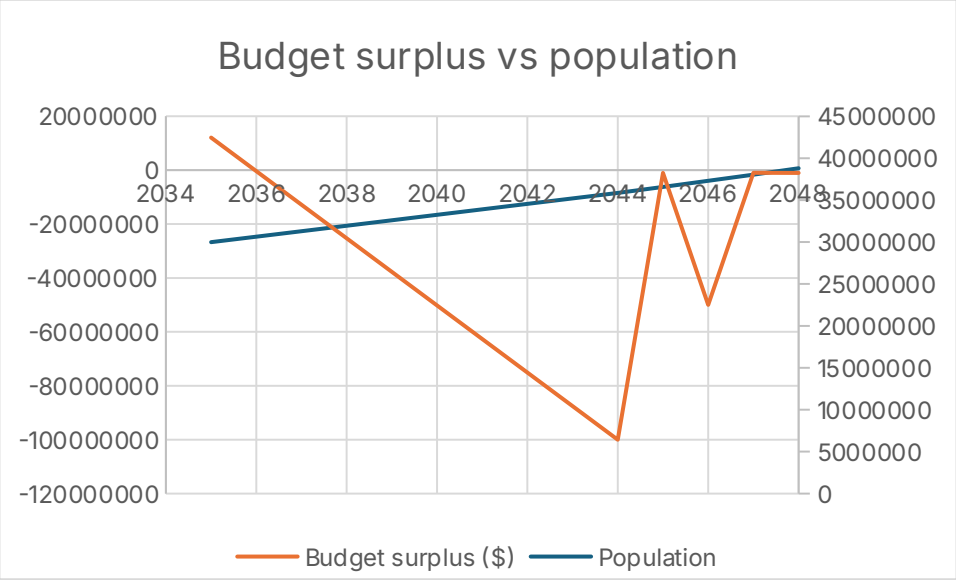


Fig.3

Party A’s Budget Mismanagement: Population Growth Strains Surplus

Part 2

Indicator 1:

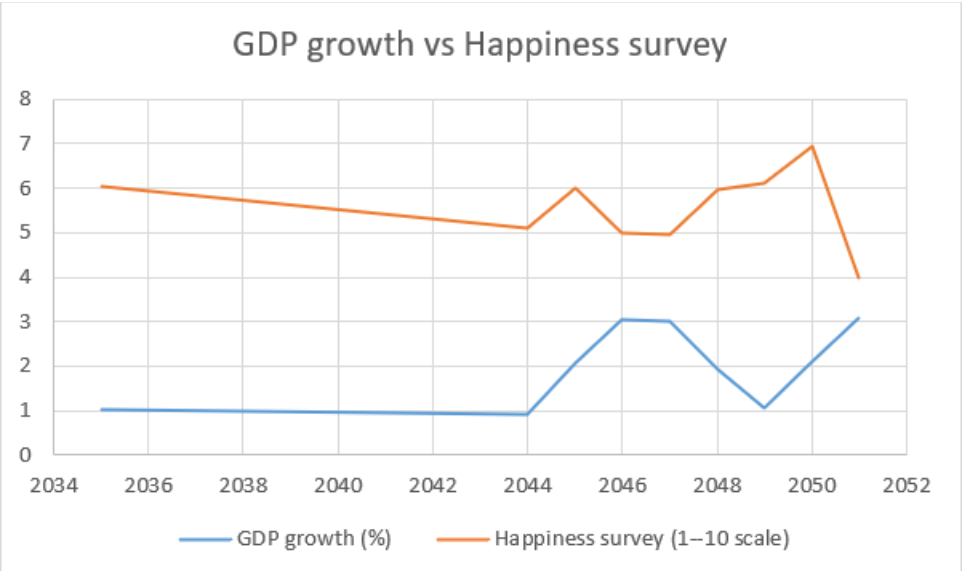


Fig. 4 Description of graph of Indicator 1.

The graph shown earlier portrayed a negative outcome because some contextual factors were not represented properly which led to the false relationship of GDP growth and happiness survey.

This [Fig.4] is the graph without any manipulation and it shows a positive correlation between GDP growth as well as Happiness rating. It is also essential to recognize that economic growth is just one of the reasons which contributes to the happiness factor, there are other factors that also affect the happiness counter, for instance, healthcare or education also have a role to play in citizen's happiness levels.

Party A was able to achieve this because they had implemented more effective economic policies that may have helped them increase the growth as well as stability which led to the data resulting in their favor.

And so to overcome this we will have to implement enhanced economic policies and focus on sustainable growth.

Indicator 2:

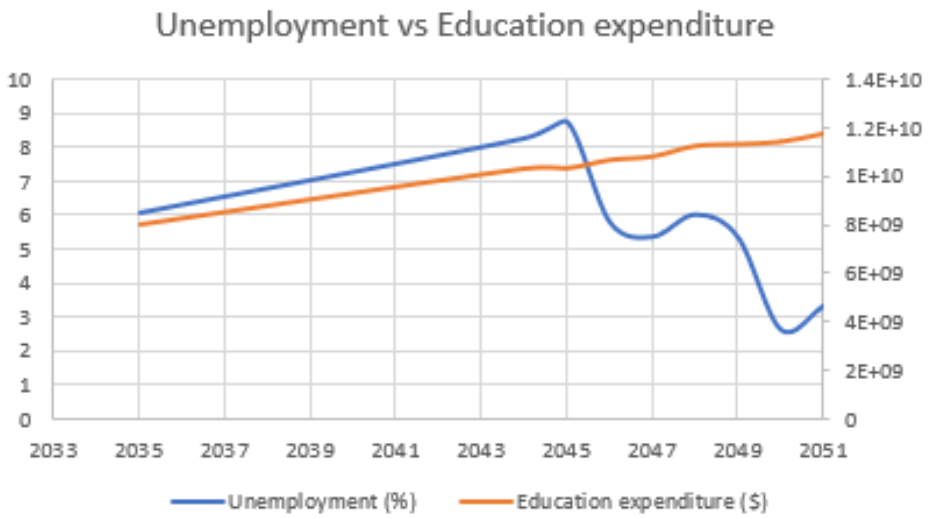


Fig.5 Indicator 2 graph representation

This is the actual analysis of the data over the years that show how unemployment rates and education expenditure acted out under the Party A rule period. As seen the unemployment rate dropped severely in the later years when Party A started increasing the education expenditure steadily.

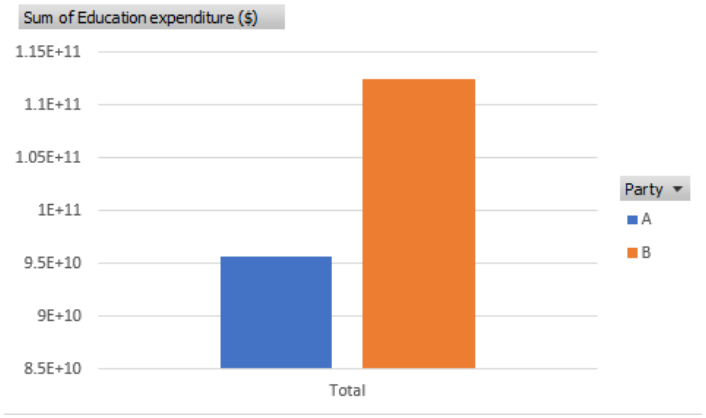


Fig.6a Education expenditure spent by both the party's

In figure 6a, It can be clearly seen that we have spent more than double what Party A spent for education, which presumably should've helped the unemployment rates go down but with the data analysis of employment during the tenure of our party vs Party A the results show something else of what was to be expected.

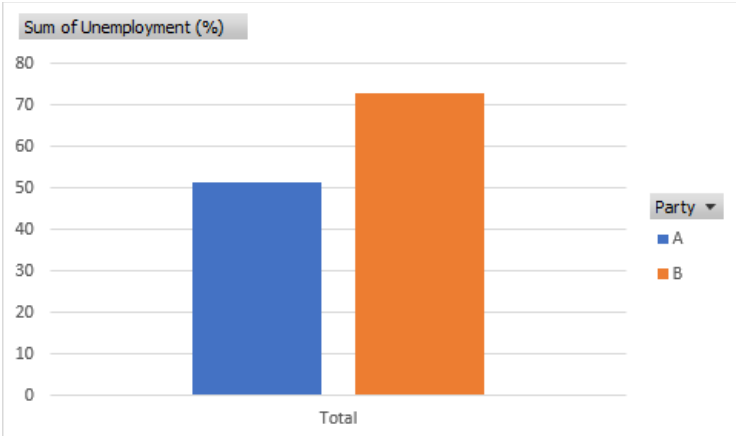


Fig.6b Unemployment rate during both the party's tenure

When we compare the unemployment rates during the tenure of both the parties Party B shows a significant amount of employment than Party A whereas Party A spent half as much on education expenditure and still they could yield more than us.

The reason why Party A could tackle the problem of unemployment and less education expenditure because of their better governance strategies which they implemented on these key indicators, now to improve our performance we need to prioritise more data-driven, decision-making and tracking the performance of the strategies implemented while making proper changes time to time.

Indicator 3

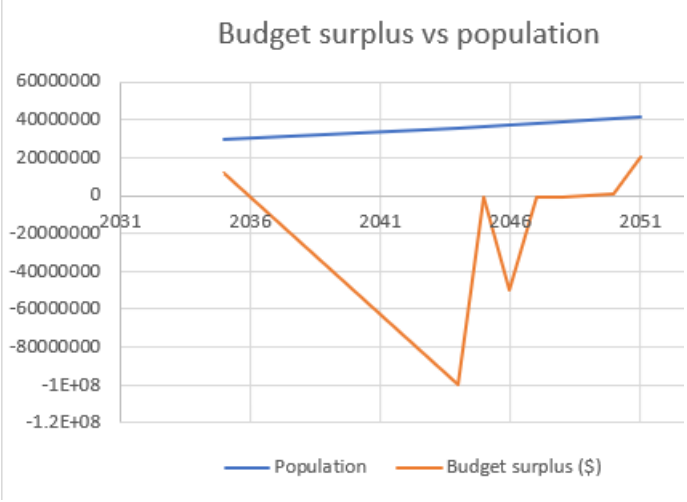


Fig. 7 Part-1, Indicator 3's actual graph

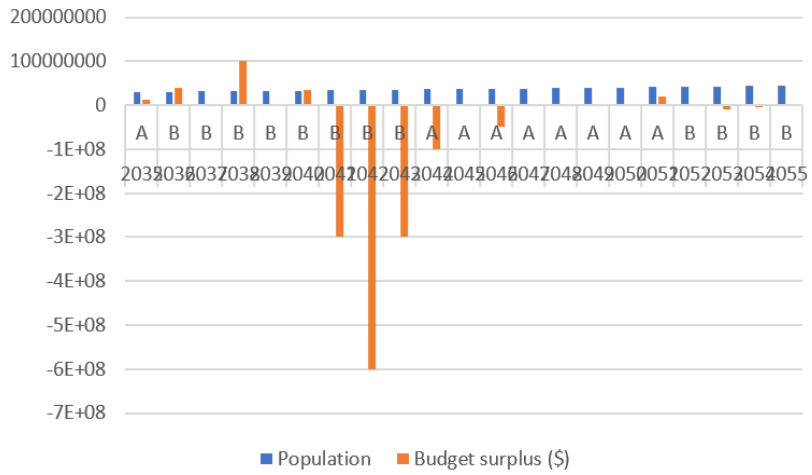


Fig. 8 Both the party's performance over the years

Both the graphs above show us that it is apparent that Party A was able to manage budget surplus more effectively, Both the parties faced challenges like negative balances in some years but Party A was able to counter it more efficiently.

To improve our party's standing we need to enhance our budget policies which should involve having clear budget guidelines and having transparency in financial decision-making processes. Moreover, we have to improve our communication strategies to convey our party's priorities and what we have to offer and state our accomplishments which will help build trust in the citizens.

Acknowledgements

I acknowledge the use of Chat-gpt in the visualization of Figures, and I used

It for producing clickbait-ey headlines.

Bibliography

The graphs are all made by the given dataset.